

Problem Statement & Objectives

Problem Statement:

Most instructors who deliver online courses today rely on a patchwork of disconnected tools — YouTube for videos, WhatsApp/Telegram groups for communication, Google Forms for quizzes, and spreadsheets for tracking students. This fragmentation makes it hard to track student progress and leaves no unified place that ties together learning content, assessment, and student management. Students, in turn, need a single place where they can browse available courses, enroll, access course content (videos, PDFs), and take quizzes/exams with automatic grading and progress tracking.

EduCore is an e-learning management system built with ASP.NET Core MVC that addresses this by providing one unified platform for two roles: **Teacher** and **Student**.

Objectives:

1. Provide a single platform that lets teachers create and manage courses, classes (units), videos, quizzes, and exams.
2. Enable students to browse a catalog of available courses, enroll, and follow content in an organized way.
3. Automate student assessment through multiple-choice quizzes and exams with automatic grading and stored attempt history.
4. Provide a secure, independent authentication system for both teachers and students, with distinct roles and permissions.
5. Build a data model that is extensible for future growth (support for Teaching Assistants, online payments, protected paid video content).
6. Give every user a personal profile experience to edit their data and change their password.